COM3031-Probabilistic Machine Learning Workshop: Laplace Approximation

Overview

This workshop mainly focuses on approximating the posterior of Bayesian logistic regression using Laplace approximation. A jupyter notebook that uses dummy data has been provided to you on ELE. You will be implementing the Laplace approximation for the posterior of Bayesian logistic regression by completing the two tasks as shown below. You may need to go through the lecture slides/notes on (Bayesian) Logistic regression and Laplace approximation. I will publish the solution in python by this week.

Laplace Approximation

In Bayesian statistics, the posterior on the parameters may not have a closed-form expression. Therefore, approximations including Laplace approximations can be used to estimate the posterior. Logistic regression is one of the well-known classification techniques. However, there is no closed-form expression on the parameters of the link function when using Bayesian logistic regression.

In this workshop,

Task 1: you will implement the Laplace approximation by (i) finding the MAP estimate and (ii) using the local curvature (Hessian) at the MAP to form a Gaussian approximation to the posterior.

Task 2: you will validate this Laplace procedure on a conjugate Beta-Binomial model by comparing the Gaussian (Laplace) approximation with the exact (closed-form) Beta posterior, and then apply the same procedure to Bayesian logistic regression to examine how different priors affect the MAP and the approximate posterior uncertainty.